

How Wind gets generated??

The most important factors are

- (i) uneven solar heating,
- (ii) the Coriolis effect due to the earth's self-rotation,
- (iii) local geographical conditions.

Only 35% of wind energy is dissipated within 1000 m of Earth's surface

Available wind energy $\sim 1.26 \times 10^9$ MW

Efficiency converting wind energy into electricity (10%) $\sim 1.26 \times 10^8$ MW

In 2025, wind supplied about 2,700 TWh of electricity, which was over 8% of world electricity.



Basics of Wind energy

Energy available in wind is basically the kinetic energy of large masses of air moving over the earth's surface. Blades of the wind turbine receive this kinetic energy, which is then transformed to mechanical or electrical forms, depending on our end use. The efficiency of converting wind to other useful energy forms greatly depends on the efficiency with which the rotor interacts with the wind stream

The kinetic energy of a stream of air with mass m and moving with a velocity V is given by

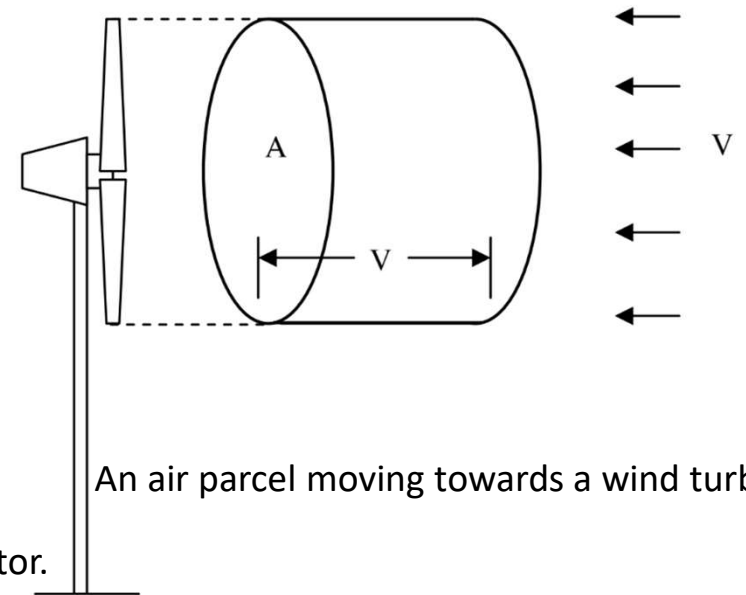
$$E = \frac{1}{2}mv^2$$

Consider a wind rotor of cross-sectional area A exposed to this wind stream

$$E = \frac{1}{2} \rho_a V^2$$

the density of air

volume of air parcel available to the rotor.



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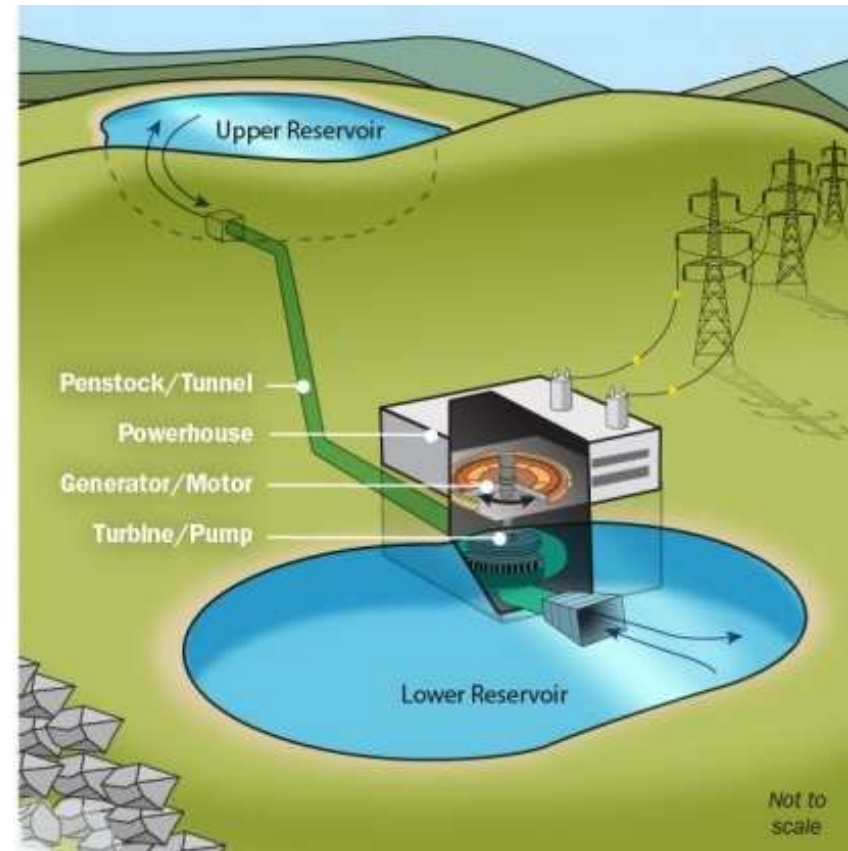
Pumped Hydro

Compared to other energy storage technologies, pumped hydro is often the most cost-effective for large-scale, long-duration energy storage.

Problem:

98% contribution is through pumped-storage hydroelectricity. it stores just 3 Wh for every 1 m³ per ton of water lifted by 1 m

Stationary energy storage solution



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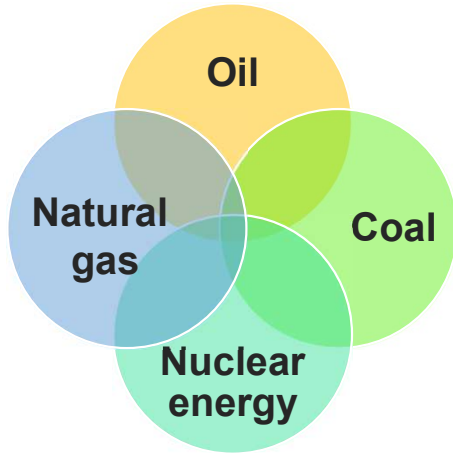
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<https://www.energy.gov/eere/water/pumped-storage-hydropower>

Traditional Energy resources

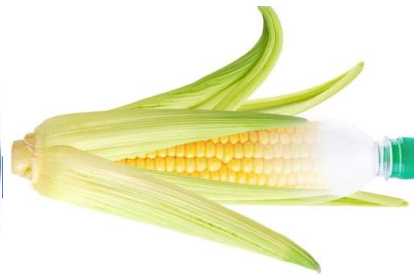
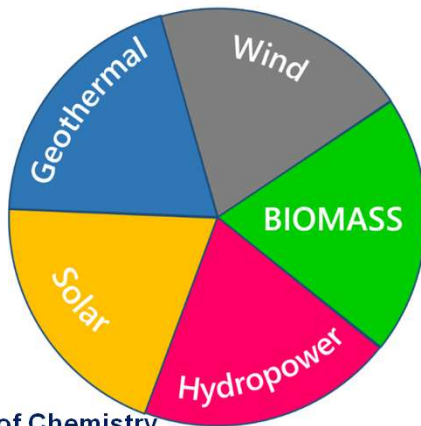


Linear economy

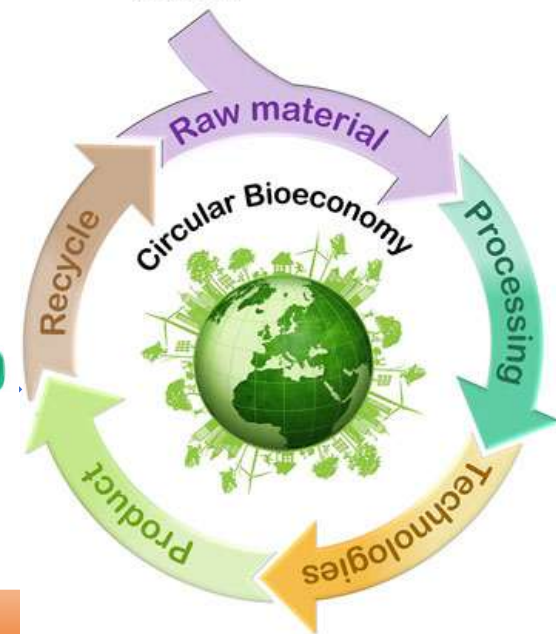


**Biomass
(Carbon)**

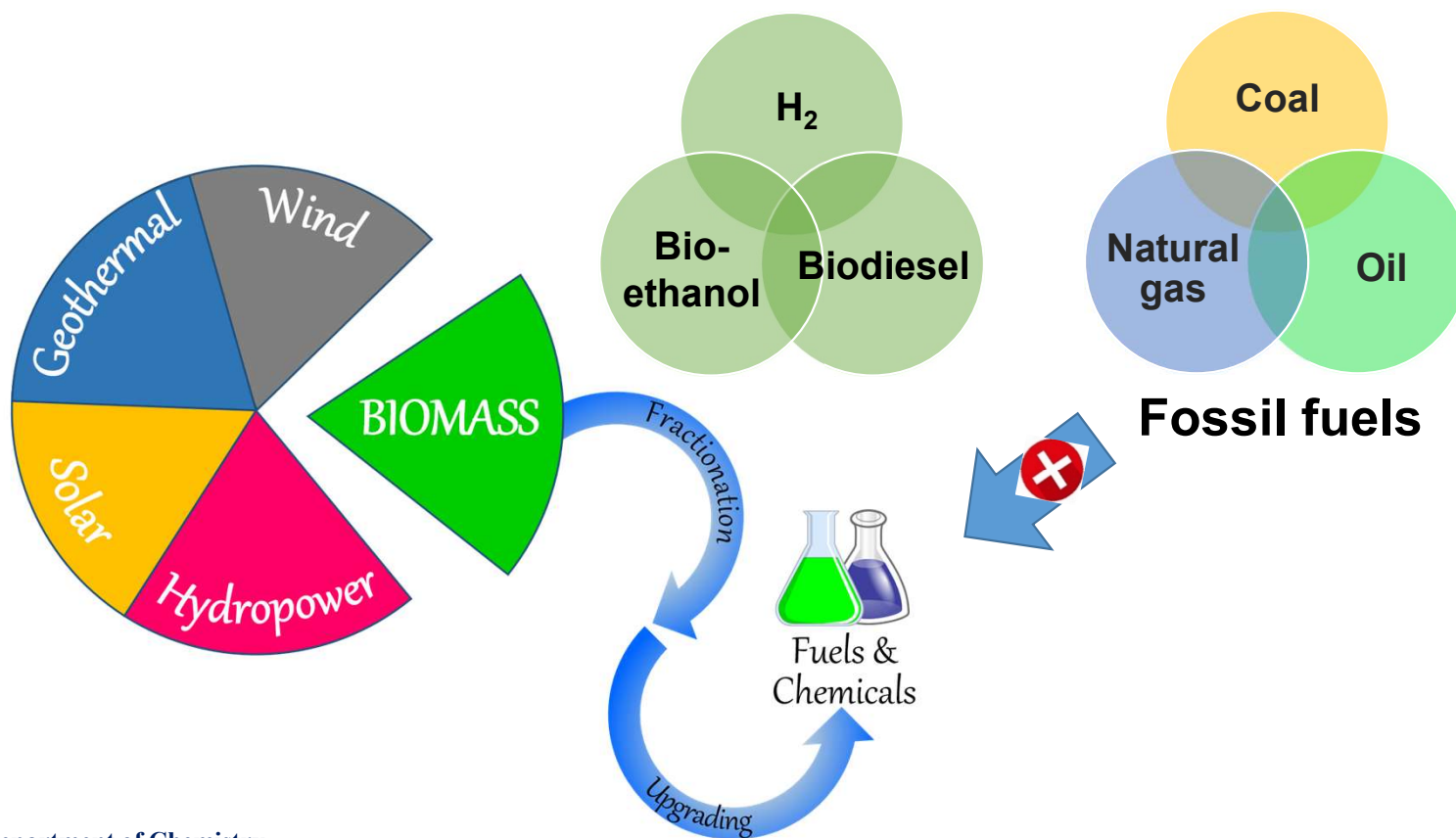
Renewables



Poly(lactic acid) bioplastic



BIOMASS – alternative to fossil fuels



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BIOMASS

- Biomass is a renewable organic material that comes from living organisms, such as plants and animals.
- **Edible biomass** (can be consumed by humans): cereals, starch, sugars, oil, crops, etc.
- **Non-edible biomass** (not suitable for human consumption): agriculture waste residues (generally lignocellulosic biomass), forest residues, and grassy materials.
- Nature itself produces huge amounts of biomass via photosynthesis using water, sunlight, and atmospheric CO_2 .
- Global annual production of biomass is estimated to be greater than 10 billion tons, equating to more than 10% of global energy supply.



Edible and non-edible biomass



Corn grains
(starch)



Corn cob
(lignocellulose)



Rice
(starch)



Rice husk
(lignocellulose)



Sugarcane juice
(sugar)

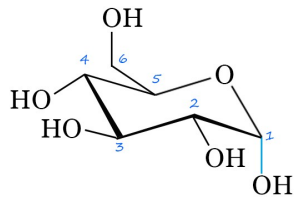
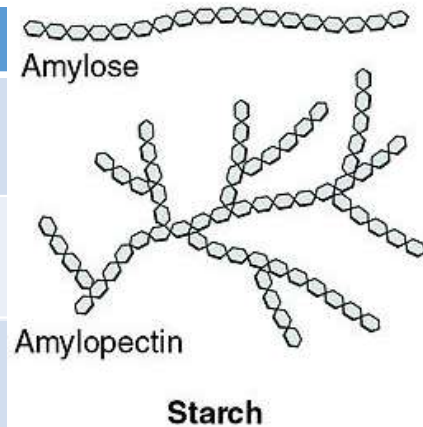


Bagasse
(lignocellulose)

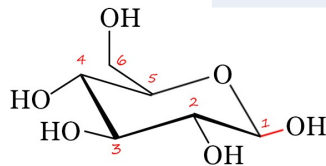
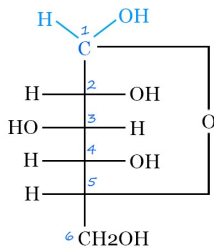


Starch and cellulose

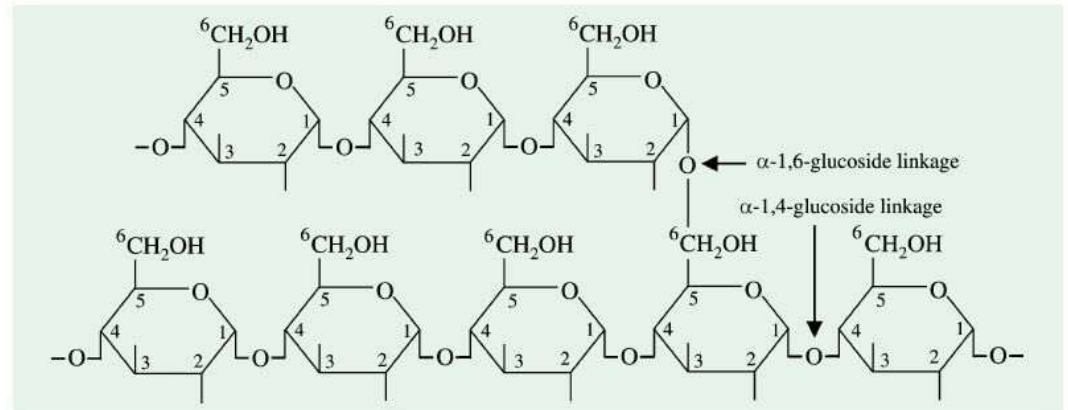
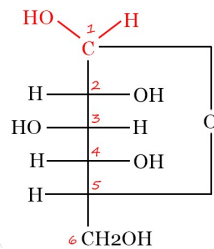
Starch	Cellulose
Glucose polymer	Glucose polymer
Linear/branched polymer	Linear polymer
Intra H-bonds	Intra and intermolecular H-bonds
Edible	Non-edible



α-D-glucose

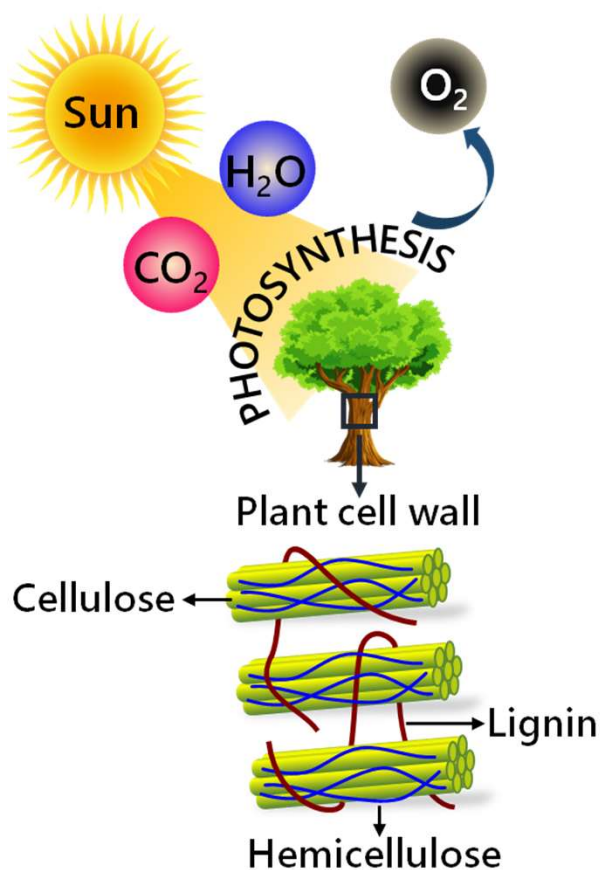


β-D-glucose



Structure of amylopectin (= B-fraction)

Lignocellulose biomass (non-edible)



Cellulose (30-50 wt%, glucose polymer)

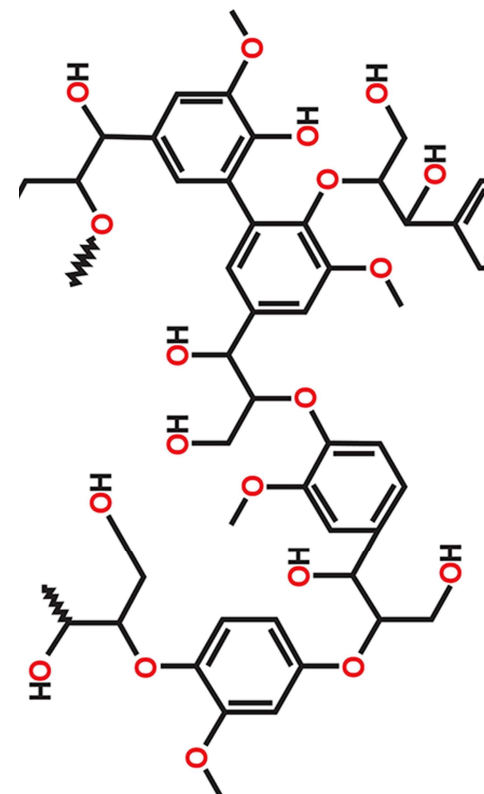
Pulp/paper production
Fuels/chemicals

Hemicellulose (20-35 wt%, polymer of C5 and C6 carbohydrates)

Improve the pulp properties
Fuels/chemicals

Lignin (15-30 wt%, aromatic polymer)

Waste from pulp industry
Aromatic chemicals



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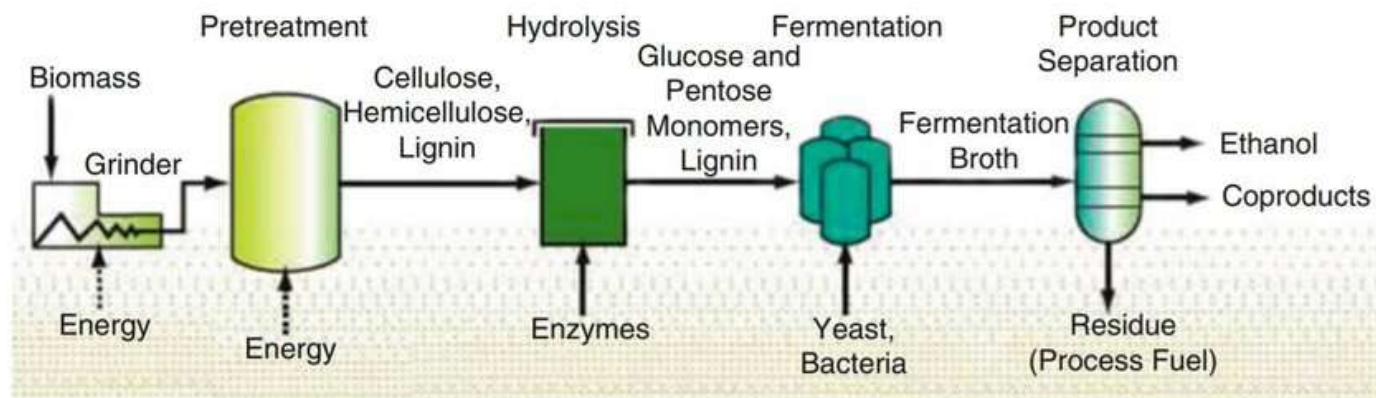
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Bio-mass to Fuels

Bio-mass can be dried and burnt for the production of heat that could be converted into electricity

The energy from bio-mass is taken by burning it directly
Or by further processing it to produce more convenient liquid and gas forms

Bio-mass can be converted into liquid forms such as Ethanol and methanol, which can be used as liquid fuel in the engines



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Bio-Gas

Bio-mass can be fermented to obtain its gaseous fuel called Bio-gas
Bio-gas has 55-65% methane, 30-40% CO₂ and rest impurities
containing H₂, H₂S and some N₂ be used as liquid fuel in the engines

Bio-Gas: The main source for production of bio-gas is wet cow dung (gobar) or wet livestock waste (or even Human waste).

The bio-gas has a particular significance in India because of large cattle population, which is about 250 million.

Some other sources of bio-gas are: Sewage, Crop residue, vegetable waste, poultry waste, algae etc.



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Geothermal Energy

This energy is embedded within the Earth. The Earth has molten core and various volcanic activities occurs in many places.

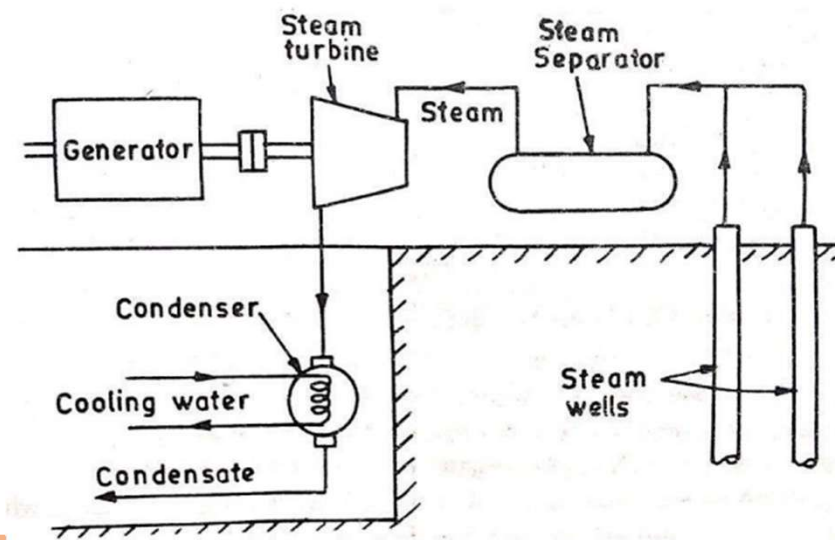
Due to the heat, the ground water is converted into steam and sometimes the hot water and steam comes out naturally due to high pressure

For large scale use, we can bore holes (steam wells) for up to 1000 m, which will release the steam and hot water with temperature up to 200-300 °C

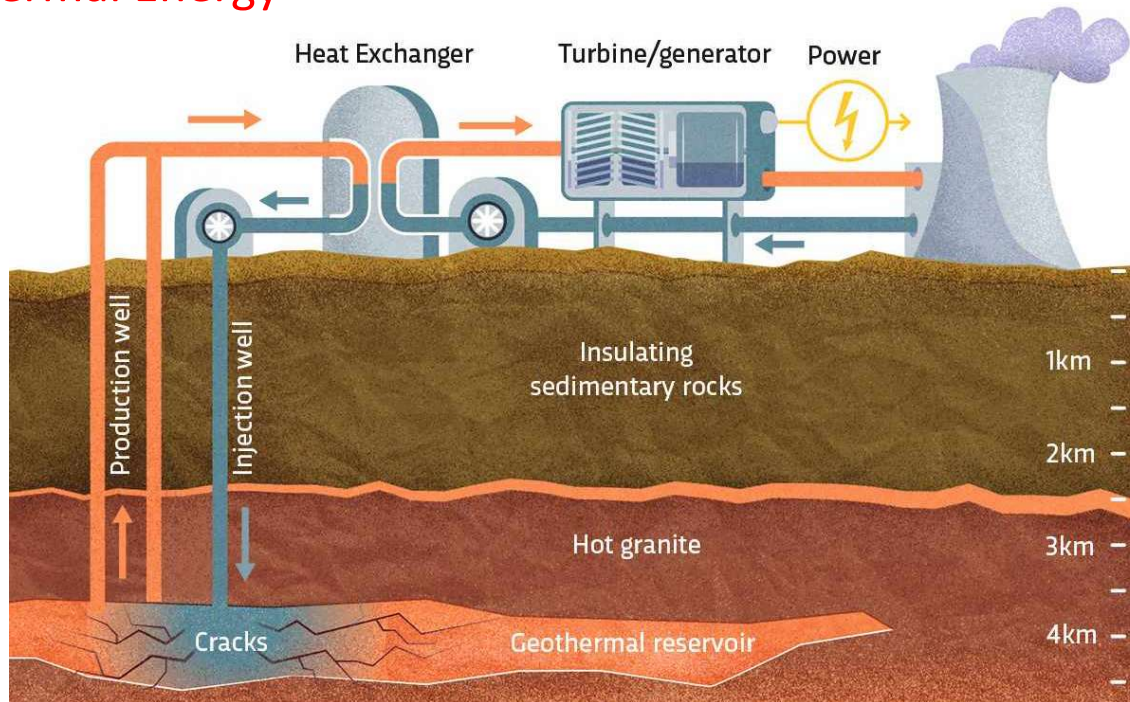
These steam can be used to operate steam turbine which will generate electricity.

The water coming out of steam wells contains high amount of dissolved minerals and the steam contains these water molecules.

So, the moisture and solid particles needs to be separated out from the steam before being fed to the steam turbine



Geothermal Energy



Advantages of Geothermal Energy

In cold countries, the geothermal energy is used directly for the melting of ice on the roads, heating houses in winters. Although the initial cost of installation is very high, the cost for maintenance and repair is negligible.



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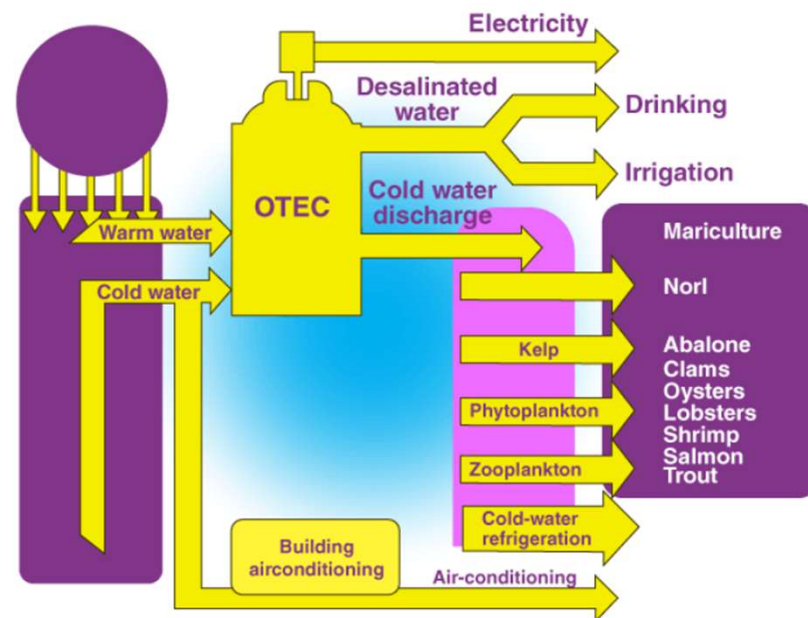
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Source: BBC Science focus magazine

Ocean Thermal Energy Conversion

- This is also as indirect source of solar energy
- The tropical oceans absorb a large amount of solar energy. The surface of oceans acts as collector of heat, while the temperature in the depths is 20-25 °C lower.
- Gradient in the temperature is used to obtain energy
- The surface water, which is heated, is used to heat some low boiling fluids such as ammonia, propane, etc
- Then the vapor produced will run the heat engine
- The exit vapor is condensed using cold ocean water of deeper regions



88000 Terawatt-hour/yr of power can be generated from OTEC

OTEC plants @ France with capacity up to 7.5 MW

In India, the department of non-conventional energy sources has proposed to set up a 1 MW OTES plant in Lakshadweep island.



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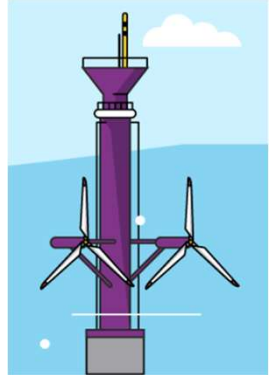
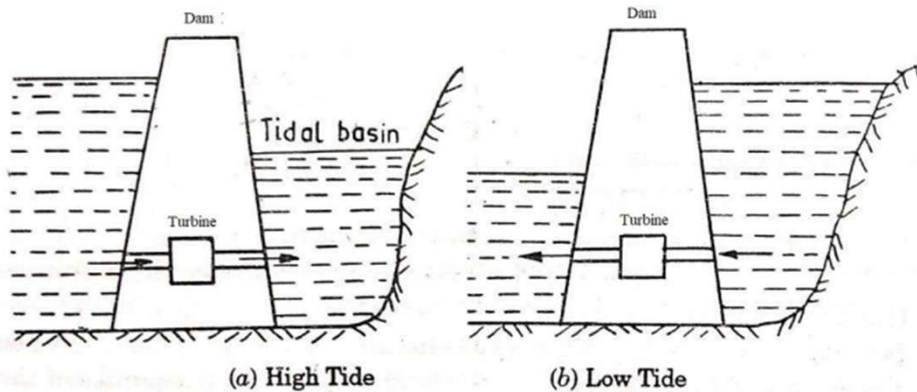
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Tidal Energy

- The tides in the sea are the result of the universal gravitational effects of the Sun and Moon on the Earth
- This causes a periodic rise and fall of the sea water level with daily rising and setting of the Sun and Moon
- These tides can be used to produce electricity which is known as tidal power
- When the water is above the mean sea level, it is called flood tide and when it is below it is called ebb tide.
- To harness the power of tides, a dam is built near the coastal area. The dam has large gates to let the water in and out. A low head hydraulic reversible turbine is used which in turn generates electricity.



[Sihwa Lake Tidal Power Station](#), located in [Gyeonggi Province](#), South Korea, is the world's largest tidal power installation, with a total power output capacity of 254 MW.

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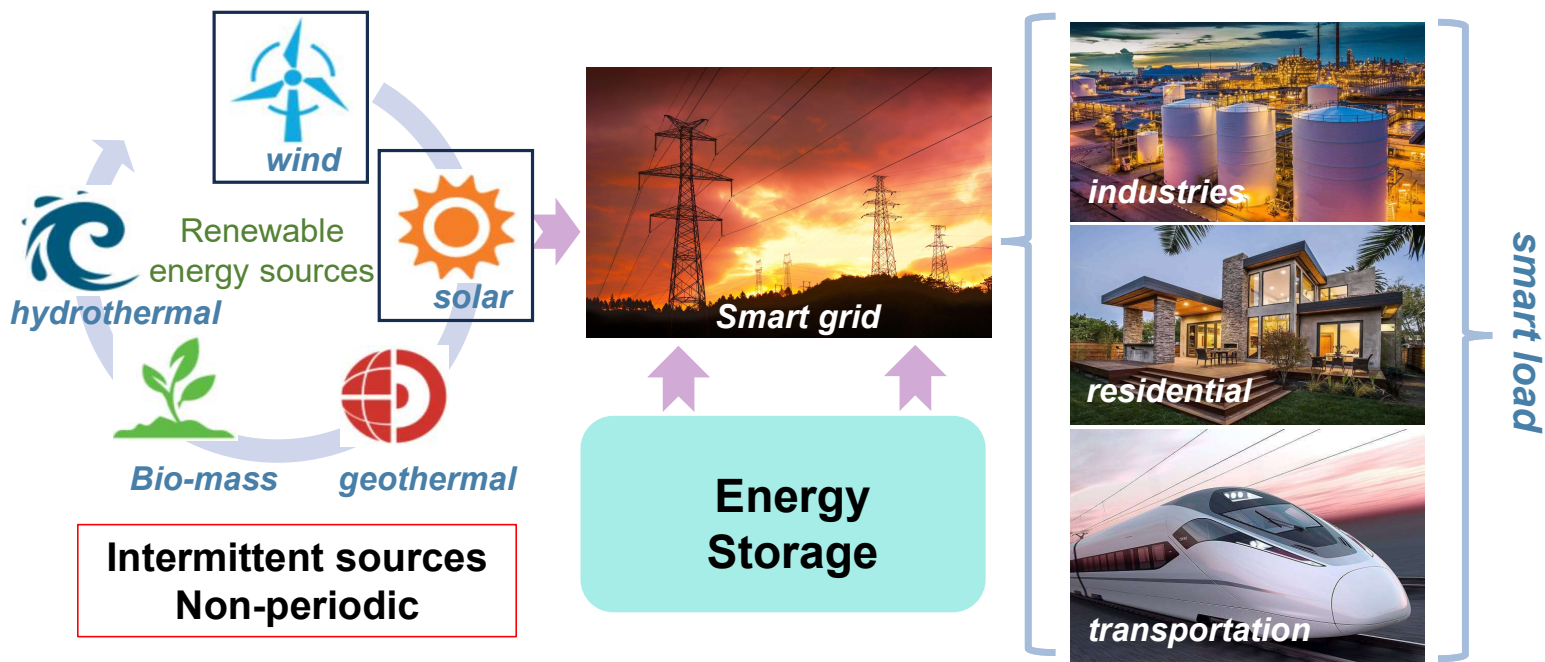
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How can we build a fossil fuel-free society?

Transition to clean energy resources



The success of renewables is governed by the potential development and integration with energy storage technologies



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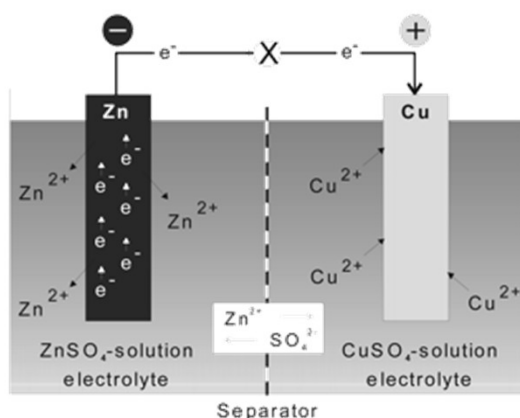
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Electrochemical cells

Energy storage/Conversion

Closed system

Battery



Anode Electrolyte Separator Cathode

Requirements on
electron conduction:
ion conduction:

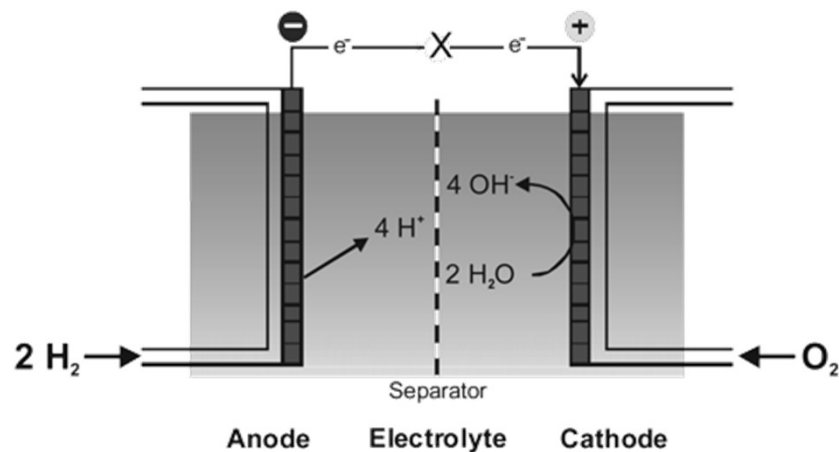
must
can

no
must

must
can

Open system

Fuel Cell



Anode Electrolyte Cathode

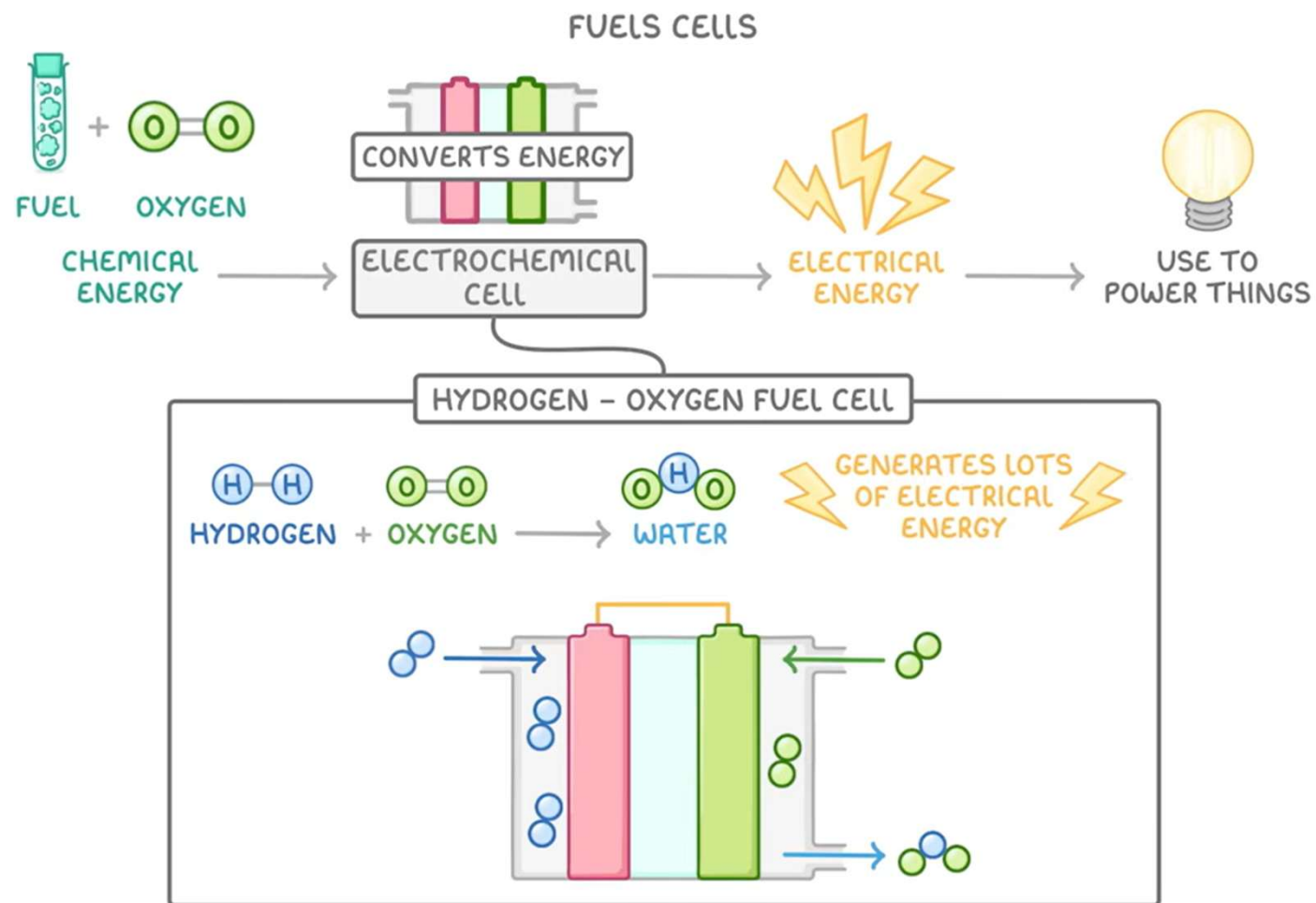


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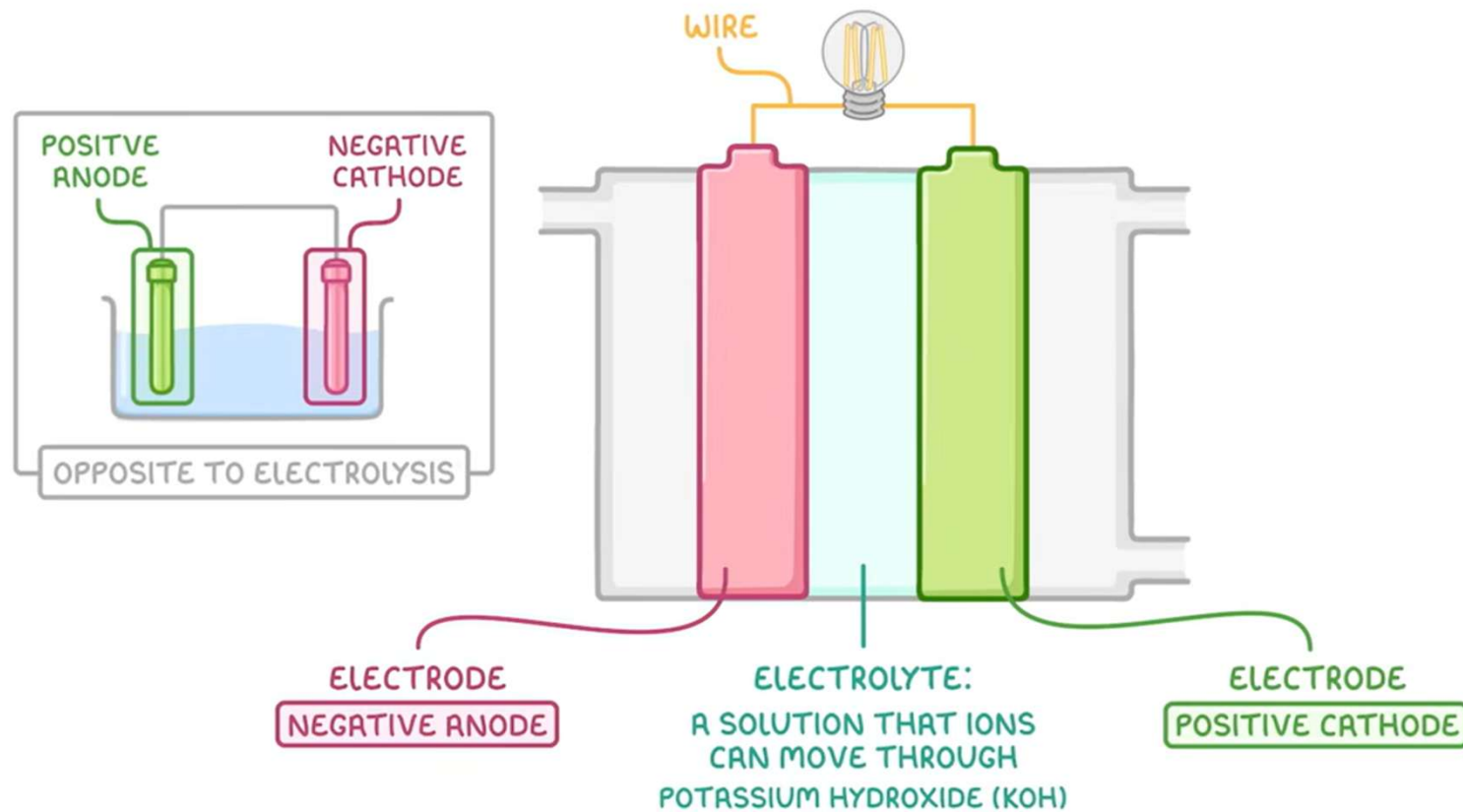
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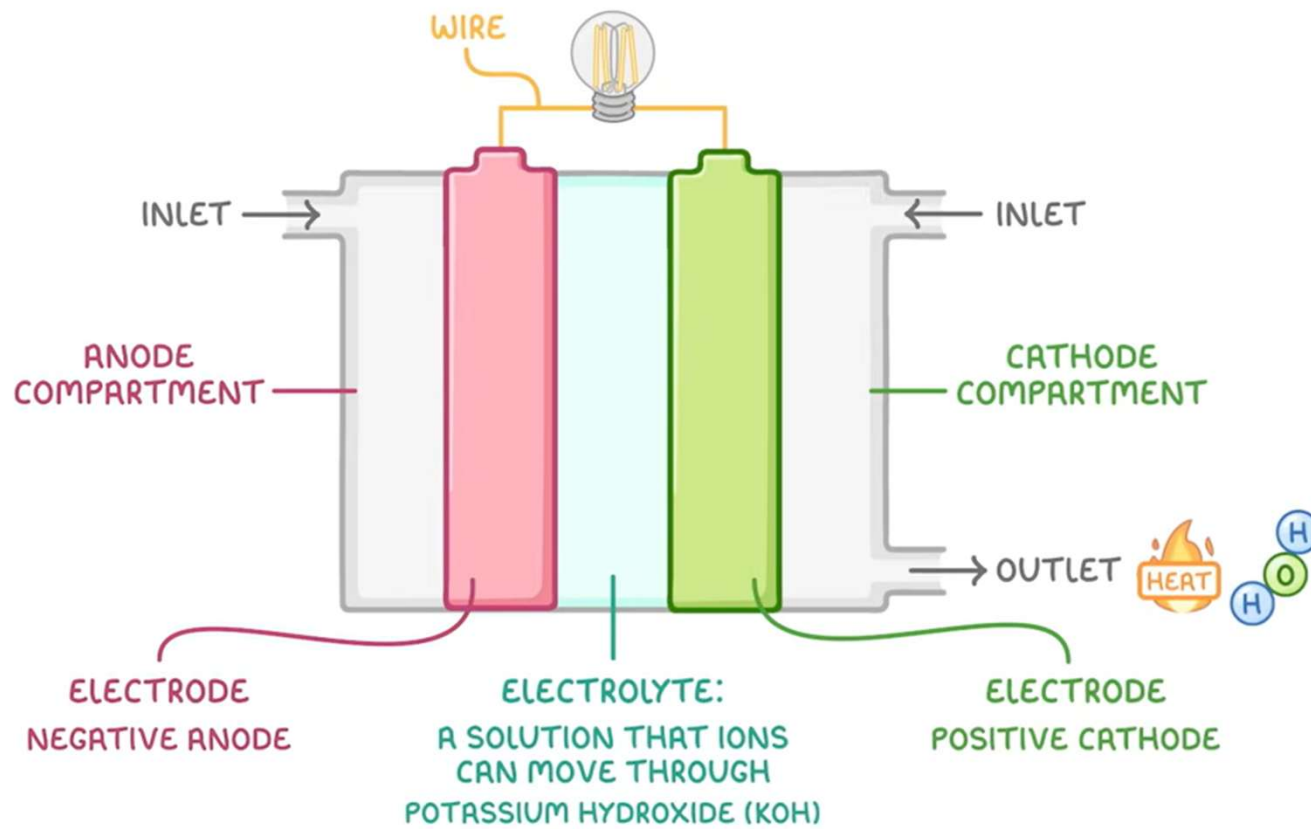
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FUEL CELL STRUCTURE



FUEL CELL STRUCTURE

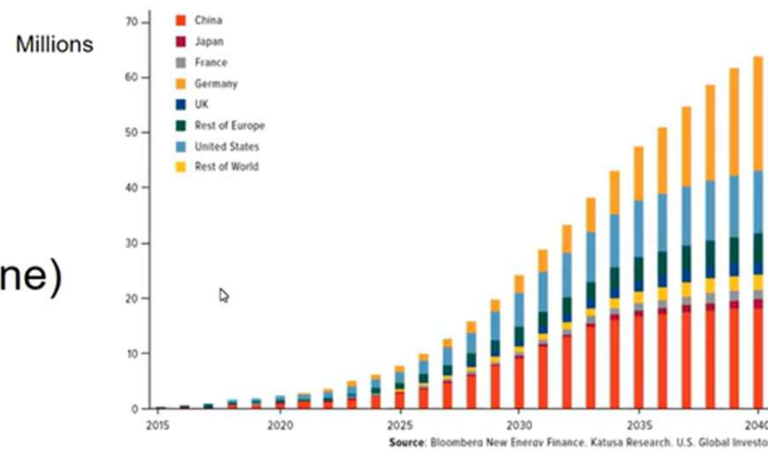


Energy Storage technologies for Electric Vehicles

Benefits

- No pollution
- No noise
- High performance
- Fast Response
- Lower driving cost (vs gasoline)
- Lower maintenance
- High safety
- Single transmission speed
- High torque and high power
- Up to 18000 rpm (6000rp for mechanical engine)
- Re-generative breaking

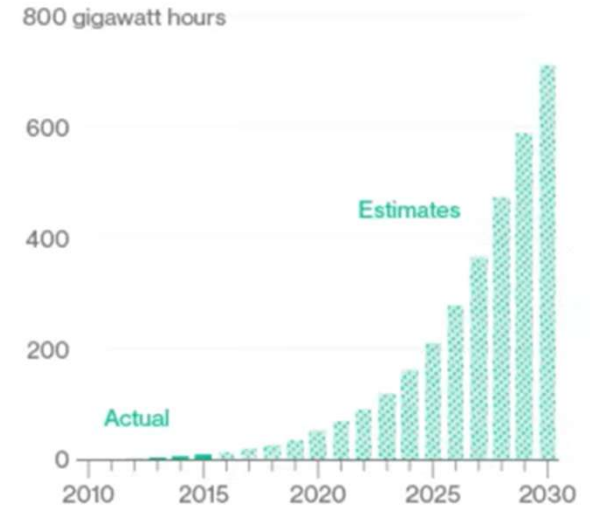
Annual Electric Vehicles Sales in Millions



Issues

- Limited range (range anxiety)
- Long charging time
- High initial cost
- Lifetime

Annual Energy Usage by EV Battery



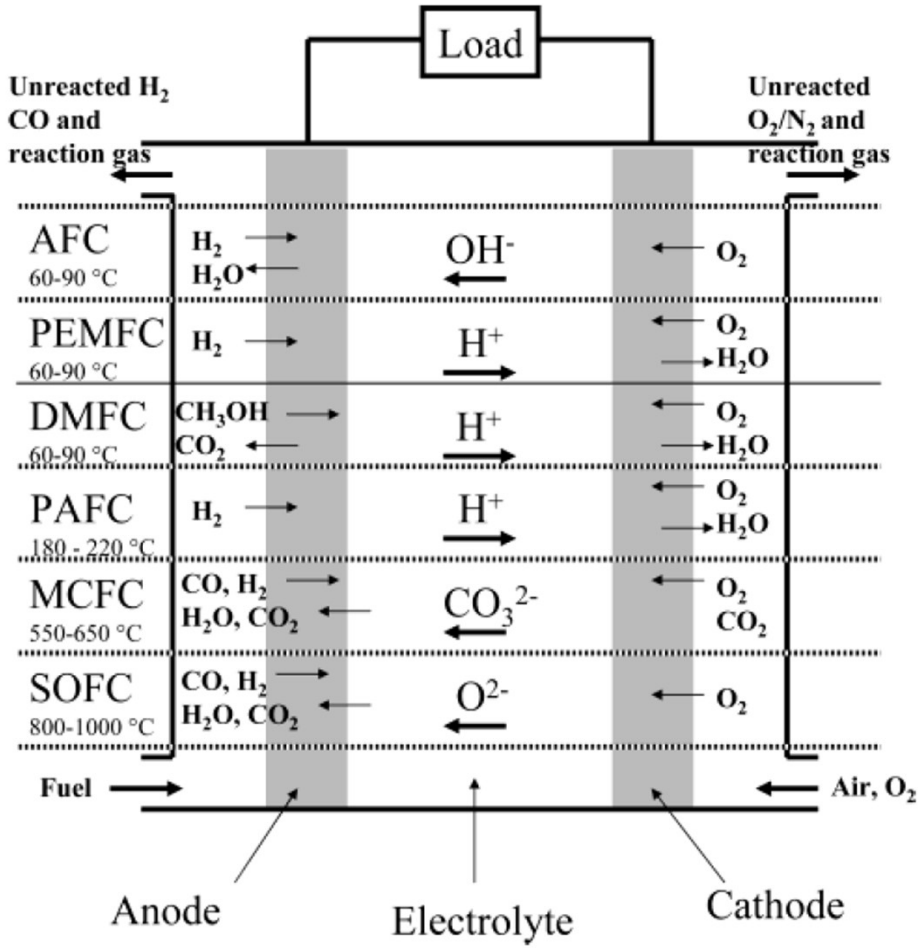
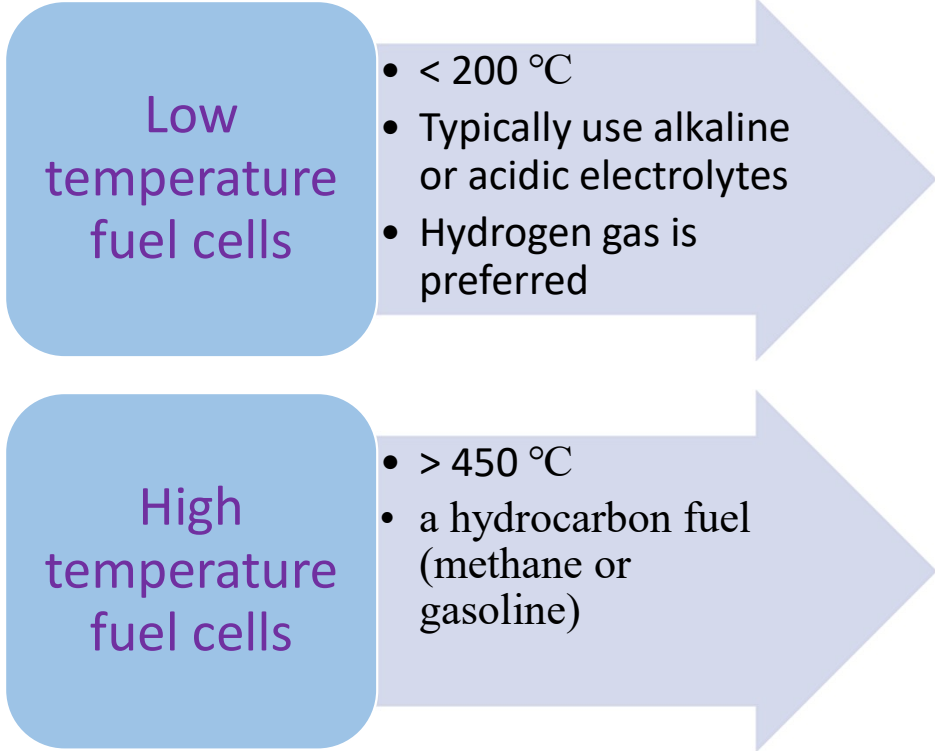
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Various types of fuel cells



The Helios unmanned aircraft utilized a hydrogen fuel cell system regenerated by solar power.



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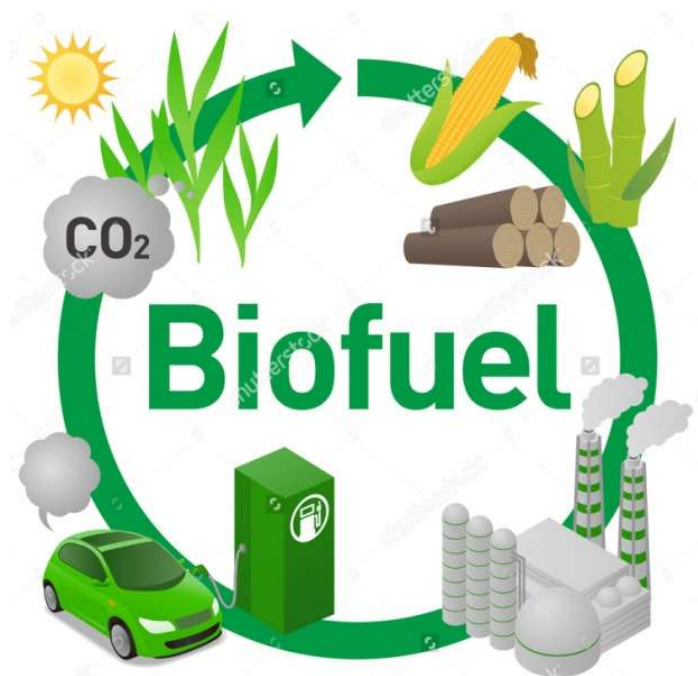
<https://wha-international.com/will-hydrogen-power-the-future-of-aerospace/>

Sustainable Aviation

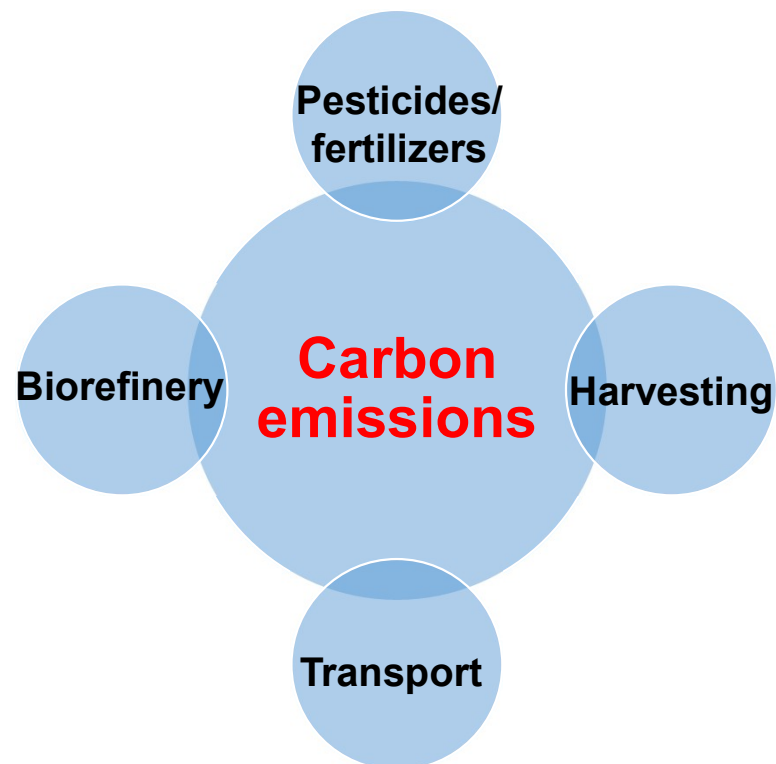


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In 2020, Airbus unveiled three concepts for hydrogen-fueled aircraft, all dubbed “ZEROe” for zero emission. They plan to launch the first craft by 2035, making it the world’s first zero emission commercial aircraft.

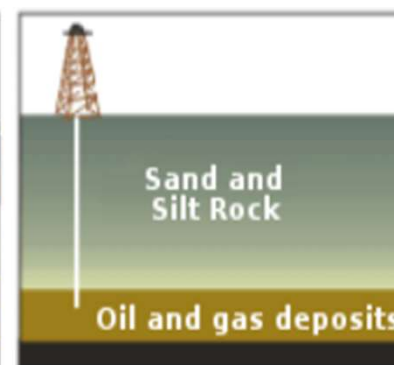
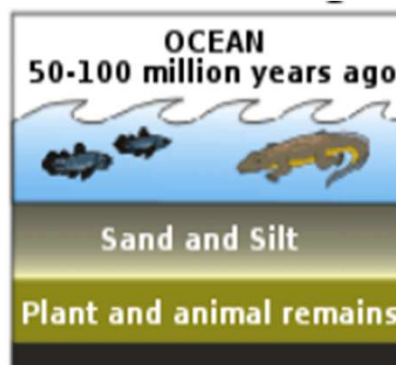
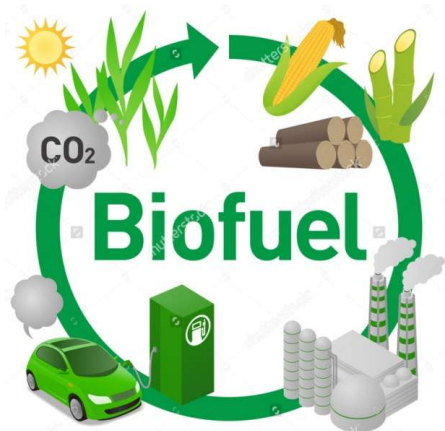


Carbon neutral



Biofuels

- ❑ Biofuels are defined as solid, liquid, or gaseous fuels derived mainly from biomass over a short-period of time. This distinguishes biofuel from fossil fuel, which is derived from long-term decomposing biological materials.
- ❑ Biofuels can lessen the dependency on fossil fuels and contribute to energy security national strategies.
- ❑ Biofuel can theoretically be produced from any biological carbon source; however, the most common sources of biofuels are photosynthetic plants and plant-derived materials.



Types of biofuels



1st Generation Biofuel

- It has **High Carbon Content**.
- Made from Edible Items. Eg- Sugar, Corn, Starch etc.

Edible biomass



2nd Generation Biofuel

- **Greenhouse Gas content less than 1st Generation Biofuel**
- Made from leftover of Food Crops. Eg- Rice Husk, Wood Chips etc.

Non-edible biomass



3rd Generation Biofuel

- It is **Carbon Neutral** in. (CO_2 Emitted = CO_2 Sequestered)
- Produced using Microorganisms. Eg. Algae

Non-edible biomass



4th Generation Biofuel

- Made from 'Genetically Engineered Crops'.
- They are **Carbon Negative**.

Genetically engineered biomass

Biofuels and carbon emissions

All **biofuels** irrespective of their source leads to lower amount of **net CO₂** **emissions** compared with **fossil fuels**

1st generation fuels (bioethanol and biodiesel):

- Source: **starch or sugarcane**
- Carbon emissions: Pesticides/fertilizers, harvesting, transport, biorefinery

2nd generation fuels (bioethanol and others):

- Source: **sawdust, corncob, bagasse, etc.** (**waste lignocellulosic biomass**)
- Carbon emissions: Transport, biorefinery (**carbon neutral**)

3rd generation fuels (biodiesel):

- Source: **Algae** (**waste biomass**)
- Carbon emissions: collecting, transport, biorefinery (**carbon neutral**)

4th generation fuels (biodiesel and bioethanol):

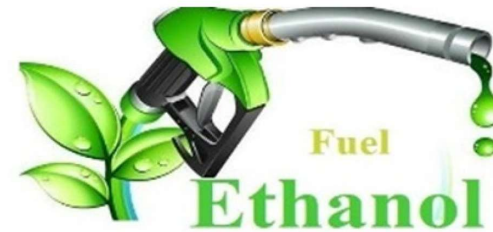
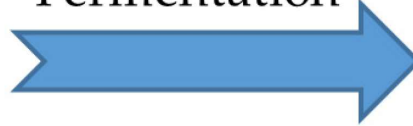
- Source: Genetically engineered microorganisms like algae and bacteria
- Carbon emissions: Pesticides/fertilizers, harvesting, transport, biorefinery

The genetic change causes the plants to absorb 30% more carbon dioxide than normal plants (carbon negative).

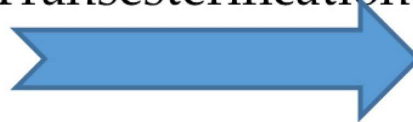
1st generation biofuels



Fermentation



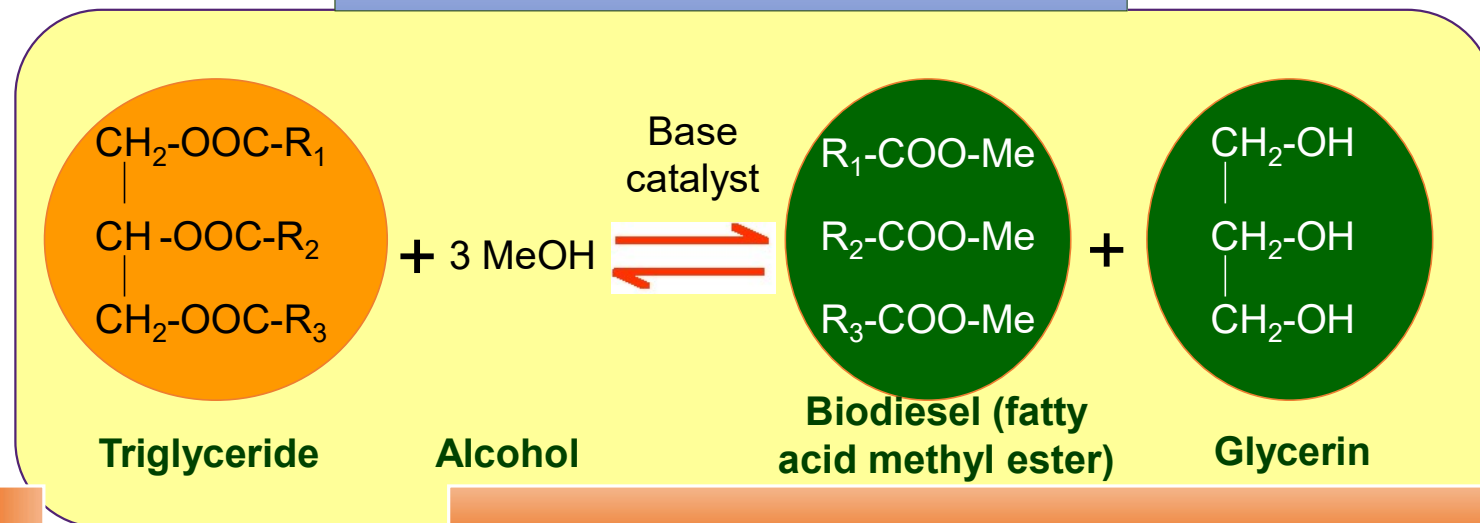
Transesterification



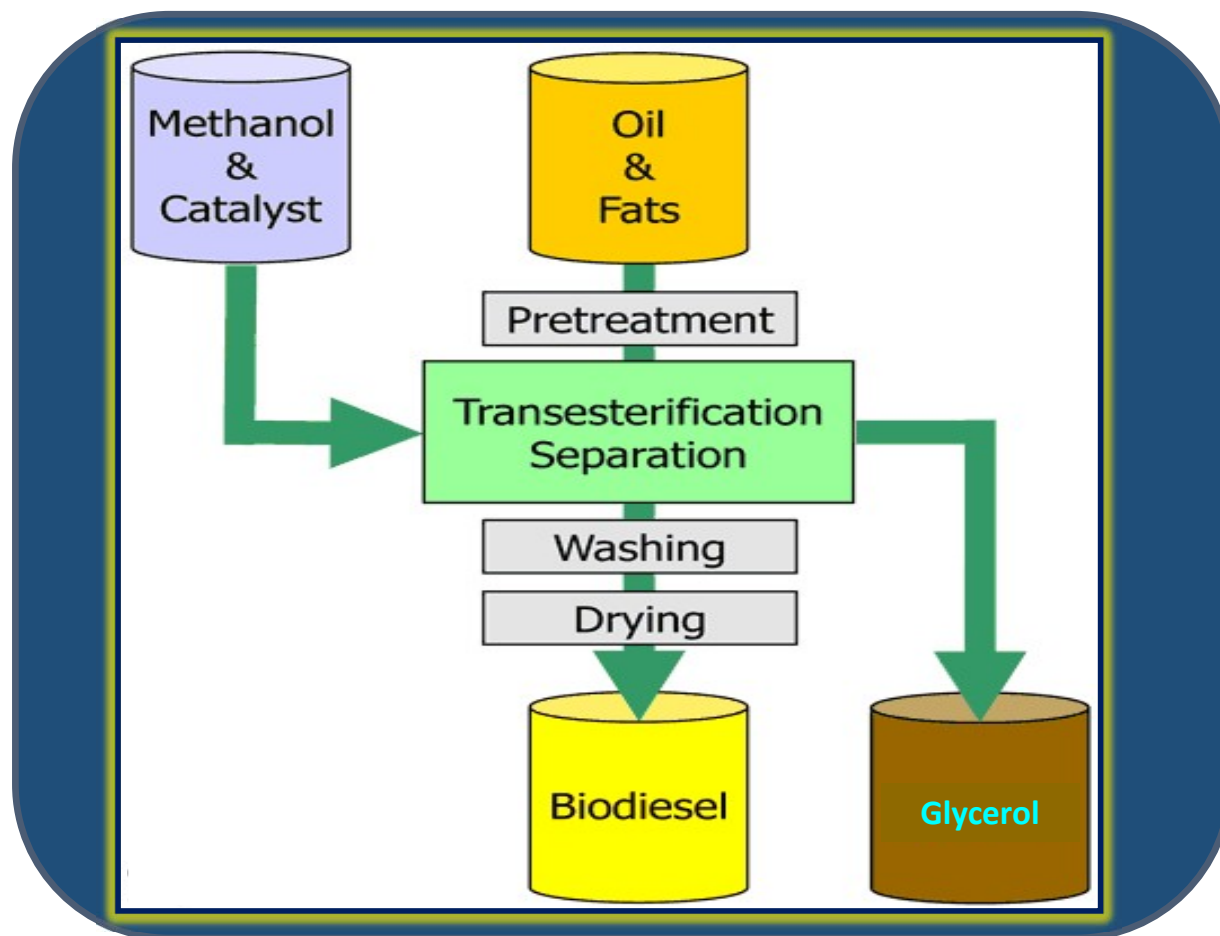
Biodiesel - Production



Transesterification Reaction



Biodiesel Production Process



Biodiesel/bioethanol blending

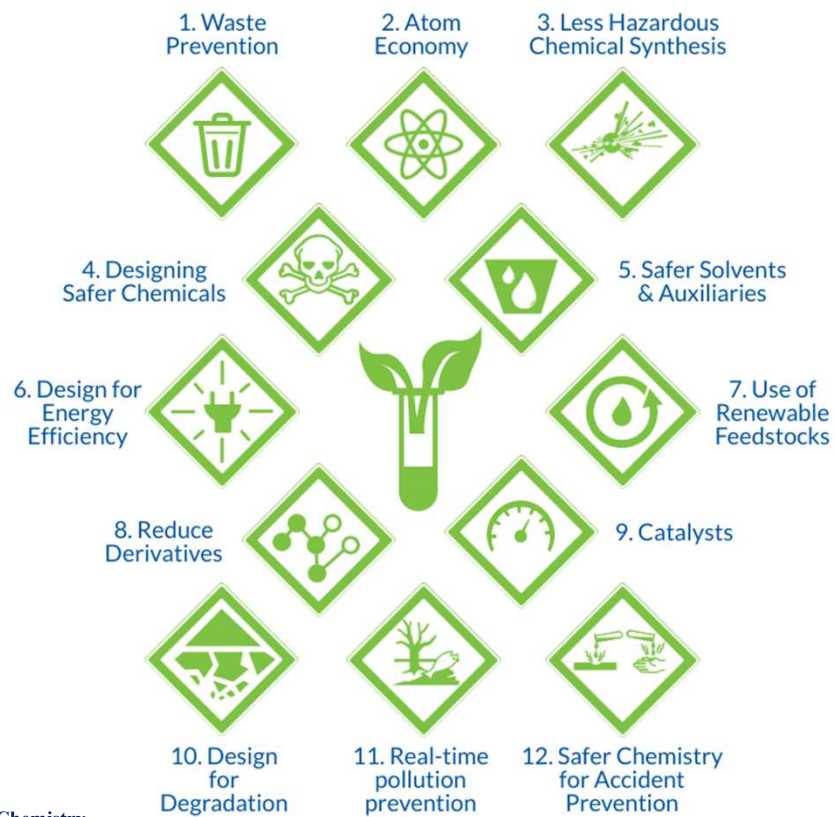
The "B" factor stands for the amount of biodiesel in fuel mix

The "E" factor stands for the amount of bioethanol in fuel mix

- B100 : 100% biodiesel
- B20 : 20% biodiesel and 80% petrodiesel
- B7 : 7% biodiesel and 93% petrodiesel
- B5 : 5% biodiesel and 95% petrodiesel
- B2 : 2% biodiesel and 98% petrodiesel
- **India (biodiesel) : B2 (B5 target by 2030)**
- **India (bioethanol) : E5 (E20 target by 2030)**

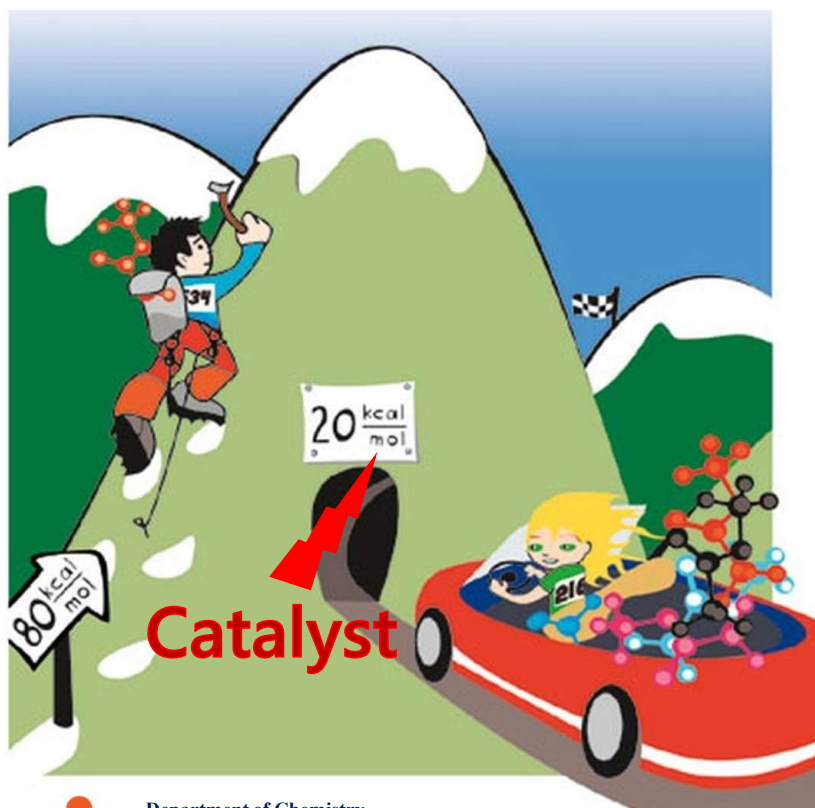
GREEN CHEMISTRY

12 principles, 1998 by Paul Anastas and John Warner



Catalysis

Catalysis is a process in which a material is used to enhance the rate of the reaction by reducing the activation energy without being consumed



Biodiesel production:

- NaOH catalyst

Steam methane reforming

- H_2 production
- Ni catalyst

Ammonia synthesis

- **Nobel prize (3 times)**
- Fe catalyst

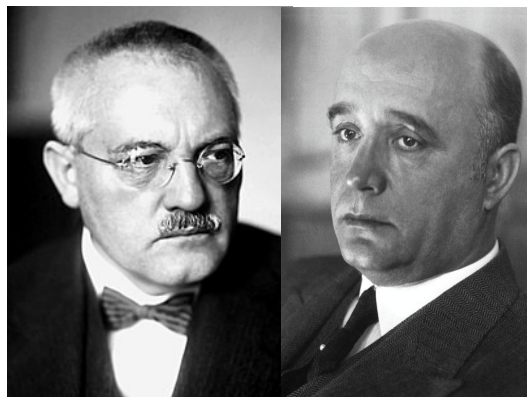
Polymer synthesis

- **Nobel prize (1963)**
- titanium tetrachloride, and an organoaluminium compound, triethylaluminium

Three Nobel prizes for ammonia synthesis



Fritz Haber
(1918)



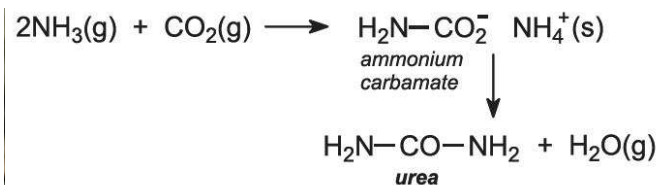
Carl Bosch **Bergius**
(1931)



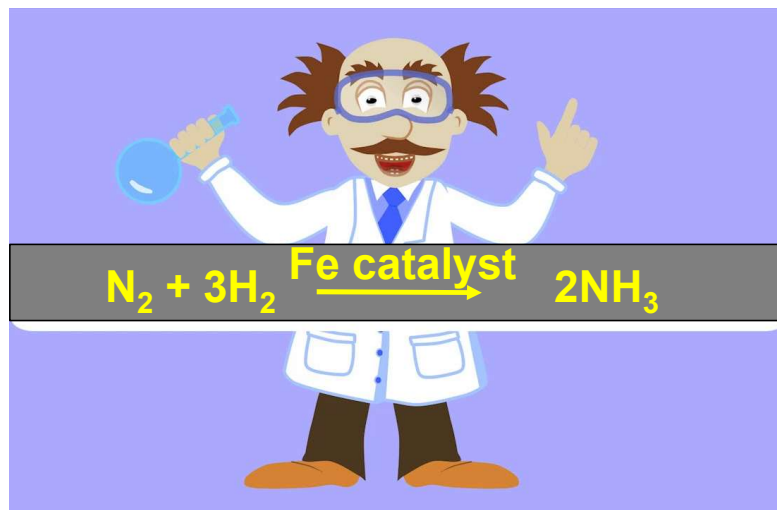
Gerhard Ertl
(2007)

Why is catalyst vital?

What is the most energy-intensive process/technology in the world?



Fertilizers
(urea, ammonium sulphate,
ammonium phosphate)



Ammonia synthesis
(Haber-Bosch process)

Ammonia synthesis is the most energy-intensive process in the world

- 1.5%–2% of global energy consumption

Catalysts and Reaction Rates

How do catalysts increase reaction rates?

- In general, catalysts operate by lowering the overall activation energy, E_a , for a reaction. (It lowers the “hill”.)
- However, catalysts can operate by increasing the number of effective collisions.
- A catalyst usually provides a completely different mechanism for the reaction.
- When a catalyst adds an intermediate, the activation energies for *both* steps must be lower than the activation energy for the uncatalyzed reaction.

Heterogeneous Catalysts

- Often we encounter a situation involving a solid catalyst in contact with gaseous reactants and gaseous products...
- Example: **catalytic converters in cars.**
 - Many industrial catalysts are heterogeneous.

How do they do their job?

- The first step is **adsorption** (the binding of reactant molecules to the catalyst surface).
- Adsorption occurs due to the high reactivity of atoms or ions on the surface of the solid.
- Molecules are adsorbed onto **active sites** on the catalyst surface.
- The number of active sites on a given amount of catalyst depends on several factors such as:
 - The nature of the catalyst.
 - How the catalyst was prepared.
 - How the catalyst was treated prior to use.

Heterogeneous Catalysts

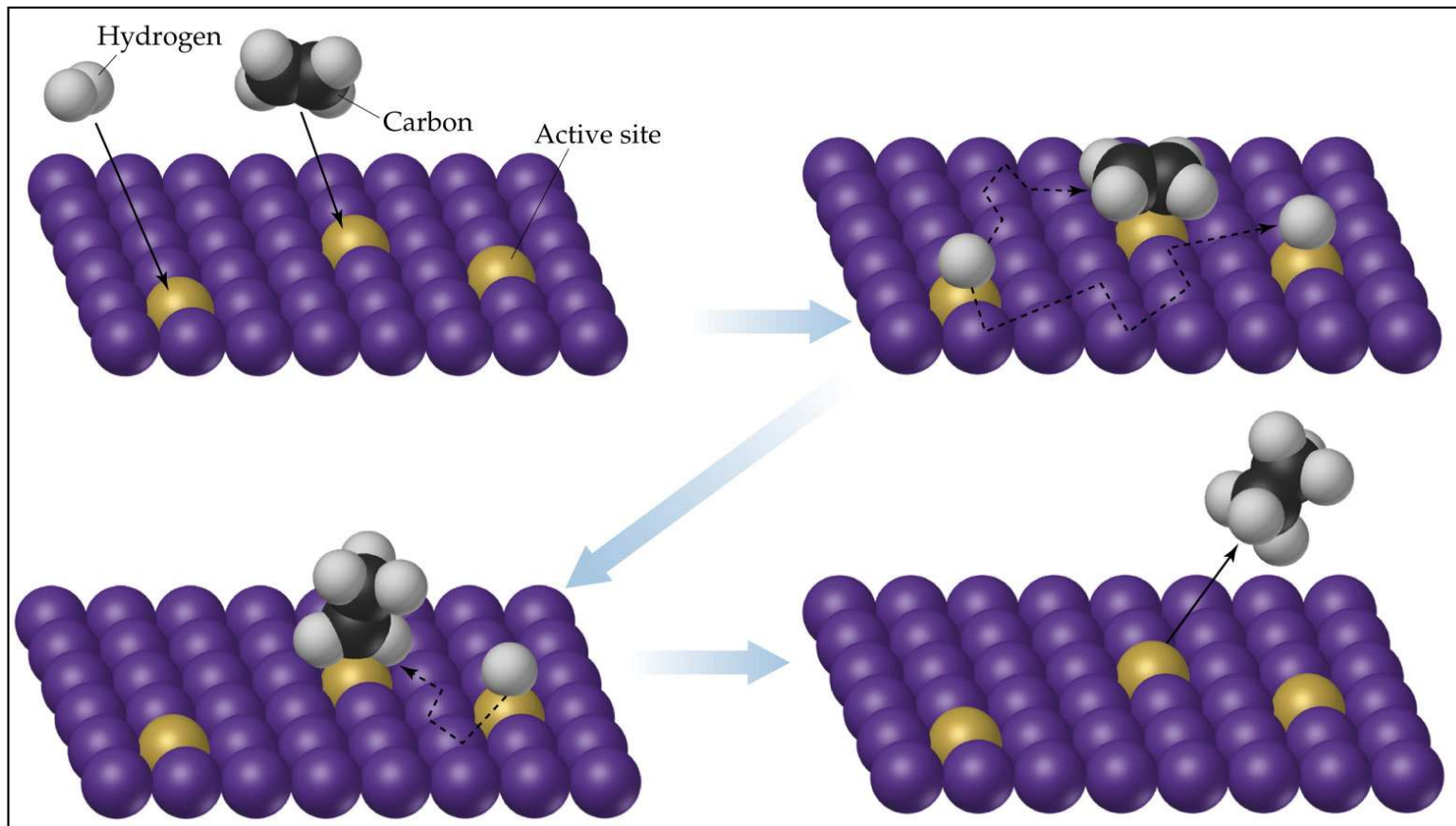
Example: $\text{C}_2\text{H}_{4(g)} + \text{H}_{2(g)} \rightarrow \text{C}_2\text{H}_{6(g)}$

- In the presence of a metal catalyst (Ni, Pt or Pd) the reaction occurs quickly at room temperature.

Here are the steps...

- First, the ethylene and hydrogen molecules are adsorbed onto active sites on the metal surface.
- Second, the H–H bond breaks and the H atoms migrate about the metal surface and runs into a C_2H_4 molecule on the surface.
- Third, when an H atom collides with a C_2H_4 molecule on the surface, the C–C π -bond breaks and a C–H σ -bond forms.
- Lastly, When C_2H_6 forms it desorbs from the surface.
- When ethylene and hydrogen are adsorbed onto a surface, less energy is required to break the bonds.
- The E_a for the reaction is lowered, thus the reaction rate increases.

Heterogeneous Catalysts



Another example:

Nitrogen gas cannot be used in the soil for plants or animals.

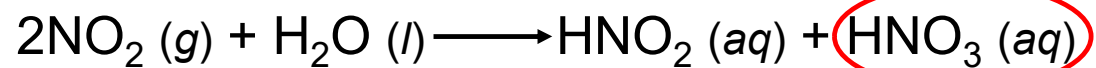
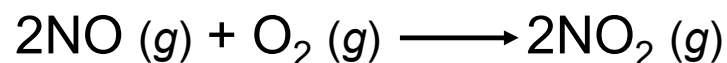
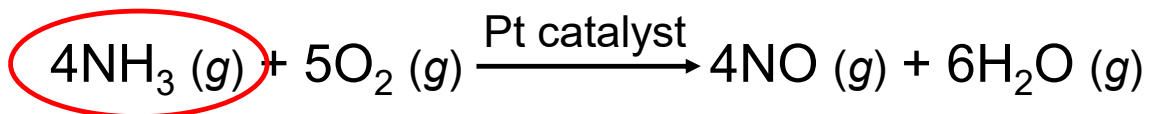
Nitrogen compounds, NH_3 , NO_2^- , and NO_3^- are used in the soil.

The conversion between N_2 and NH_3 is a process with a high activation energy (the $\text{N}\equiv\text{N}$ triple bond needs to be broken).

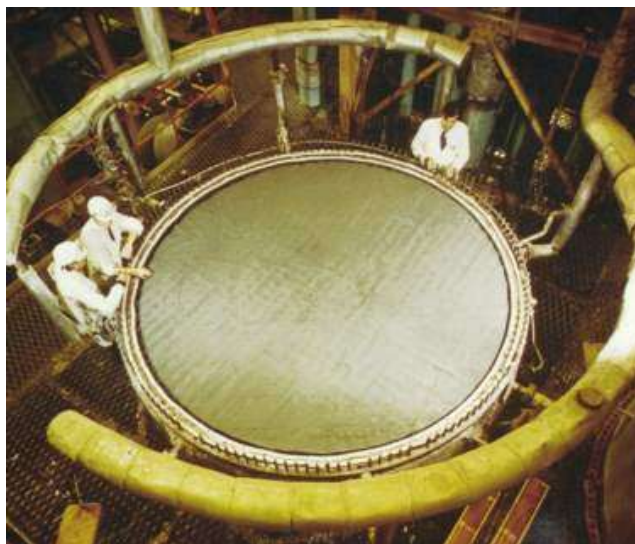
Nitrogenase, an enzyme in bacteria that lives in root nodules of legumes such as clover and alfalfa, catalyzes the reduction of nitrogen to ammonia. It lowers the E_a , and the reaction proceeds.

Ostwald Process

Ammonia



Nitric acid



Pt-Rh catalysts used
in Ostwald process

Catalytic Converters

